

JEFFREY BLAY

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PROFESSIONAL SUMMARY

Geospatial data scientist and spatial data engineer completing a Ph.D. in Geospatial Data Science (Dec 2026), with 5+ years building machine learning systems, ETL pipelines, GIS applications, and spatial analyses over large-scale geospatial data. Combines deep learning (PyTorch, CNNs/transformers) with GIS development (ArcGIS, QGIS, web mapping) and cloud deployment, and a record of turning complex spatial data into decision-ready products for government, research, and community stakeholders. Nine peer-reviewed publications and an open-source Python library on PyPI.

PROFESSIONAL EXPERIENCE

Geospatial Research Engineer

Jun 2026 – Present

Remote Sensing & GIS Lab, North Carolina A&T State University — Greensboro, NC

- Developing a QGIS plugin that packages CNN and transformer-based flood-depth models for on-demand inference, making research-grade models usable by non-technical GIS end users.
- Assessing disaster impacts on crop-specific land cover and vulnerable communities using federal datasets and Google Earth Engine to support resilience and risk-mitigation planning.
- Leading a peer-reviewed journal publication on a physics-informed AI framework for urban flood modeling and infrastructure vulnerability assessment.

Geospatial Data Science Research Assistant — NASA Environmental Data Project

Sep 2024 – May 2026

GEMS Institute, North Carolina A&T State University — Greensboro, NC

- Developed and optimized CNN and transformer segmentation models (Swin-UNet, SegFormer, UNet, Attention UNet, UNet++) in PyTorch for urban flood-depth prediction, achieving ~0.19 ft average MAE.
- Designed a physics-informed deep learning framework embedding domain-knowledge constraints, improving physical consistency of predictions by ~84% over conventional deep learning baselines.
- Built geospatial ETL pipelines in Python (rasterio, GDAL, pandas) to ingest, align, and preprocess ~347 million pixel-level records from multi-source raster data for model training.
- Published ezprocess, an open-source Python library (PyPI/GitHub) that cut geospatial data-preparation time by ~85% across machine learning workflows.
- Led a 4-person research team to design and publish a novel flood-depth benchmark dataset for reproducible urban inundation modeling.
- Managed code, experiments, and dependency versioning in Git/GitHub, enabling collaborative development across 2 research teams.

Graduate Research Assistant — NSF Multimodal Data Fusion Project

Sep 2023 – Aug 2024

Remote Sensing & GIS Lab, North Carolina A&T State University — Greensboro, NC

- Built ETL workflows in Python (NumPy, pandas) preparing ~700K geospatial point records for flood prediction models, reducing data preparation time by 60%.
- Trained and evaluated Random Forest and XGBoost models using train-test splits, cross-validation, and performance metrics to assess predictive accuracy and reliability.
- Developed automated geospatial QA/QC and validation tests verifying spatial accuracy, attribute consistency, and data integrity prior to public release.
- Designed automated multi-sensor preprocessing pipelines with ArcPy in ArcGIS Pro for a 5-member team publishing a multi-source geospatial dataset.
- Curated ~1,000 web images into a labeled training dataset via Python web scraping (BeautifulSoup, SQLite), and built a PyTorch framework to train and evaluate GoogLeNet and ResNet18 classifiers for automated flood detection in crowdsourced imagery.

GIS & Community Engagement Manager

May – Aug 2023

Urban Resources Initiative — New Haven, CT

- Managed 7 community stakeholder groups, translating competing priorities into weekly project plans and implementation strategies.
- Delivered ArcGIS Pro and QGIS analyses, maps, and visualizations enabling stakeholders to evaluate site-specific options and refine proposed investments.

Research Assistant — NASA Environmental Justice Project

Sep 2022 – May 2023

Hixon Center for Urban Ecology, Yale University — New Haven, CT

- Analyzed multispectral satellite imagery at census-block level to identify communities vulnerable to urban heat island effects, building ArcPy and SQL workflows for land surface temperature data.
- Ran time-series analysis in R (dplyr, ggplot2) to characterize heat and tree-cover disparities across urban communities.

Research Fellow — TRI Research Fellowship

May 2022 – May 2023

Tropical Resources Institute, Yale University — New Haven, CT

- Trained a U-Net regression model on Landsat imagery to predict building composition across metropolitan Ghana, achieving R^2 of ~0.75 on held-out test data.
- Processed VIIRS nighttime-lights data in Google Earth Engine and applied geostatistical analysis to map electricity infrastructure gaps across built-up areas.
- Deployed an integrated field mapping and monitoring system (ArcGIS Pro, ArcGIS Online, Field Maps, Dashboards) capturing 300+ ground-truth observations and improving collection accuracy by 90%.

Geospatial Data Analyst

Sep 2020 – Jun 2021

National Census Secretariat, Ghana Statistical Service — Accra, Ghana

- Performed enterprise-scale QA/QC on national census datasets, validating coordinate reference systems, boundary alignment, geometry, and attribute integrity across point, line, and polygon data.
- Restructured Supervisory Area boundary datasets, resolving spatial errors, misaligned boundaries, and redundant features to improve national data consistency.
- Produced near-real-time coverage analytics with a 15-member team and identified difficult-to-enumerate areas to direct field resource allocation during census operations.

TECHNICAL SKILLS

Programming: Python, R, SQL, JavaScript
Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Random Forest, CNNs & transformers (UNet, Attention UNet, Swin-UNet, SegFormer), physics-informed neural networks, model evaluation (RMSE, MAE, SSIM, AUC-ROC), RAG / LLM APIs
Geospatial & Remote Sensing: ArcGIS Pro, ArcGIS Online, ArcGIS Dashboards & StoryMaps, ArcPy, QGIS (plugin development), Google Earth Engine, GDAL/OGR, GeoPandas, rasterio, PostGIS, ENVI, spatial data QA/QC
Data Engineering & Databases: ETL pipeline design, Apache Airflow, web scraping, PostgreSQL, MySQL, SQLite, IBM DB2
Statistics & Analytics: Spatial statistics, geostatistics, time-series analysis, multivariate analysis, hypothesis testing
Cloud, DevOps & Web: AWS (S3, EC2), Docker, CI/CD, Git, GitHub Actions, HPC, Render, FastAPI, REST APIs, Leaflet.js, MapLibre GL, Streamlit
Visualization: Matplotlib, Seaborn, Power BI, Tableau, R Shiny

SELECTED PROJECTS

GridVision MLOps
Python · FastAPI · PostGIS · Supabase · GitHub Actions · MapLibre GL · Render

- End-to-end MLOps system forecasting U.S. electricity demand with live data ingestion, automated retraining, metric-gated model promotion to production, and interactive 3D visualization.

jeffreyblay.github.io/energy_demand_mlops

Quickview Geodata Portal
Python · FastAPI · GeoPandas · Leaflet.js · Render · GitHub Pages

- Open-source web platform for fast, interactive spatial data exploration of multiple geospatial file formats loaded directly from URLs.

jeffreyblay.github.io/quickview-geodata-portal

Weather Alert Dashboard
Python · FastAPI · JavaScript · Docker · CI/CD · REST API · GitHub Pages

- Real-time weather monitoring platform integrating the National Weather Service API, with REST filtering, analytics, and risk scoring across all 50 U.S. states.

jeffreyblay.github.io/climate-alert-dashboard

EDUCATION

Ph.D., Applied Science & Technology — Geospatial Data Science, North Carolina A&T State University
M.E.S., Environmental Science — Environmental Data Science for Urban Sustainability, Yale University
B.A., Geography with Political Science, University of Ghana (GPA 3.81/4.00)
Certifications: IBM Data Engineering Professional Certificate, Coursera (in progress, exp. Sep 2026); Foundations of AI Engineering, CodePath (2026)

Expected Dec 2026
May 2023
May 2019

PUBLICATIONS & HONORS

9 peer-reviewed journal and conference publications (5 first-author), including Remote Sensing, Data in Brief, Scientific Data, IEEE Access, and IGARSS; 11 conference presentations at IGARSS, AGU, ISPRS Geospatial Week, and ASPRS.
Honors: Outstanding Doctoral Graduate Research Assistant Award, NC A&T College of Science & Technology (2026); Lidar Leader Award — Best Poster, ASPRS & Geo Week Conference (2024)